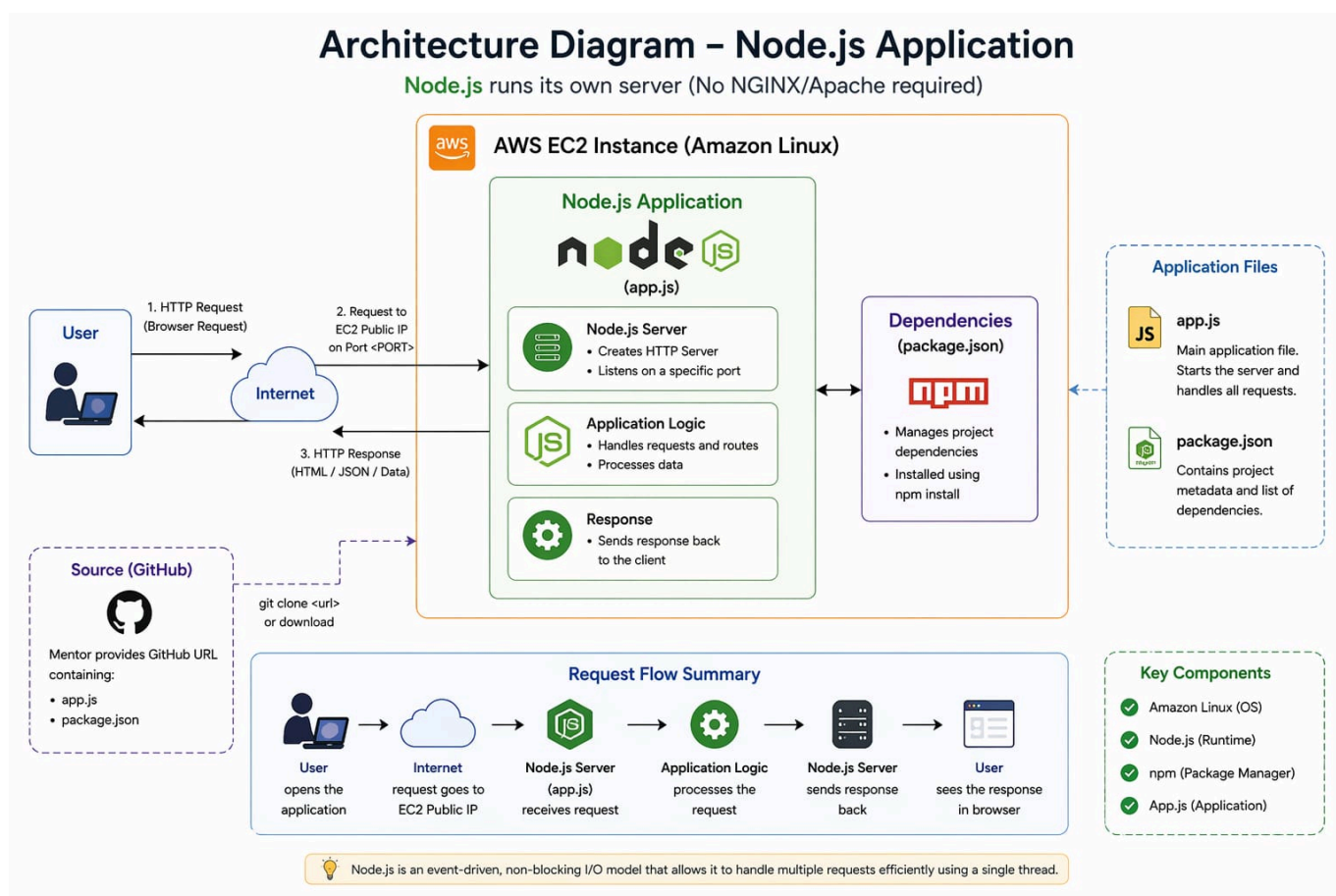


Node.js Application Deployment on Amazon Linux (EC2)

Introduction

This is my another project where I deployed a Node.js application on an Amazon Linux EC2 server. In this project, I learned how Node.js works as its own server, so no external web server like NGINX or Apache is required. The application files (app.js and package.json) were provided by my mentor through a GitHub URL, and I deployed the project by cloning and running it on the server

Architecture



Technologies Used

- Amazon Linux (EC2)
- Node.js
- NPM (Node Package Manager)
- Git

Features

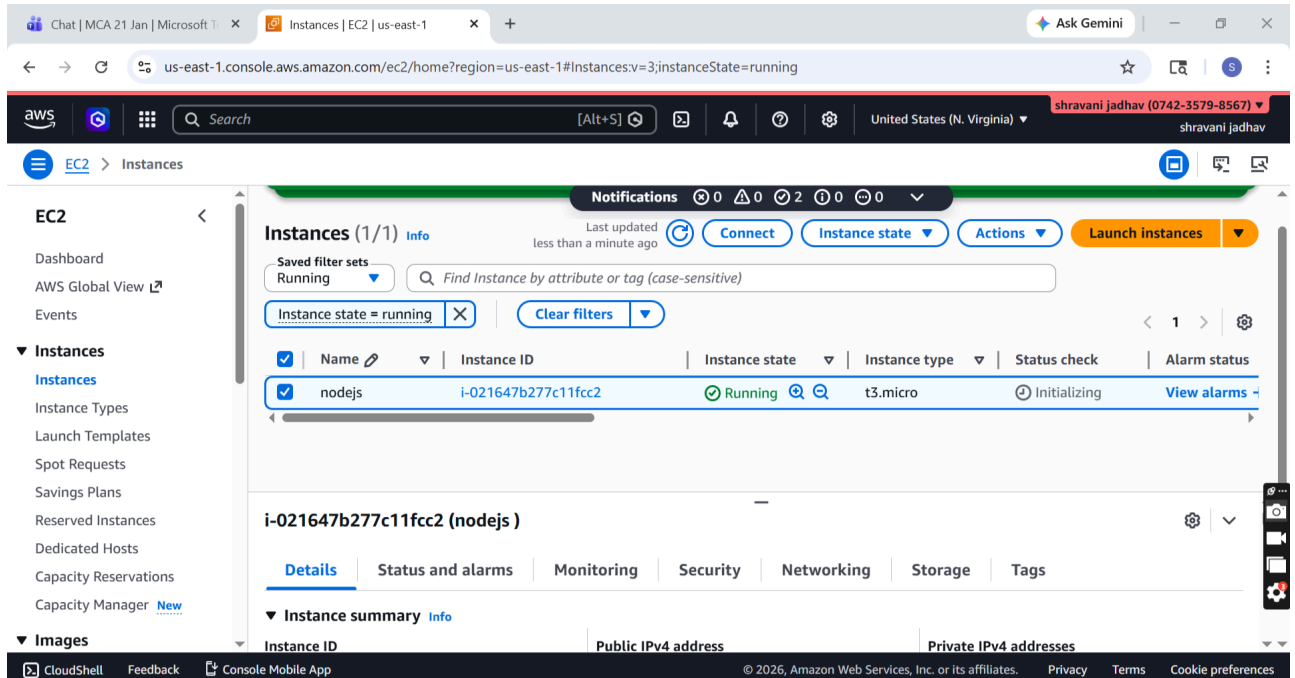
- Node.js runs its own server
- No need for NGINX or Apache
- Application runs using node app.js

- Accessible via EC2 public IP and port

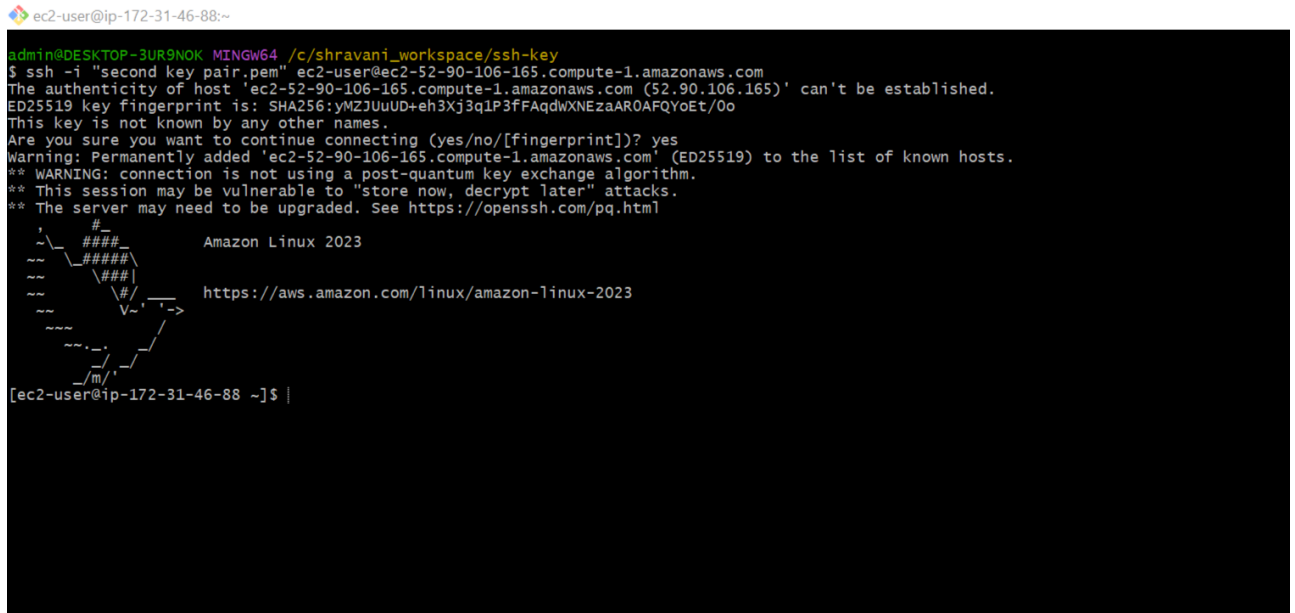
Setup & Deployment Steps

1. Launch EC2 instance

- Create Amazon linux EC2 instance



- Connect using SSH



2. Install Git

```
sudo yum update
sudo yum install git -y
```

```
ec2-user@ip-172-31-46-88:~  
[ec2-user@ip-172-31-46-88 ~]$ sudo yum update  
Amazon Linux 2023 Kernel Livepatch repository                               259 kB/s | 31 kB    00:00  
Last metadata expiration check: 0:00:01 ago on Wed Apr 22 18:18:36 2026.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[ec2-user@ip-172-31-46-88 ~]$ sudo yum install git -y  
Last metadata expiration check: 0:00:18 ago on Wed Apr 22 18:18:36 2026.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
git	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	53 k
Installing dependencies:				
git-core	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	4.9 M
git-core-doc	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	2.8 M
perl-Error	noarch	1:0.17030-2.amzn2023.0.1	amazonlinux	42 k
perl-File-Find	noarch	1.37-477.amzn2023.0.7	amazonlinux	25 k
perl-Git	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	41 k
perl-TermReadKey	x86_64	2.38-9.amzn2023.0.2	amazonlinux	36 k
perl-lib	x86_64	0.65-477.amzn2023.0.7	amazonlinux	15 k

```
=====
```

Transaction Summary				
Install 8 Packages				
Total download size: 7.9 M				
Installed size: 41 M				
Downloading Packages:				
(1/8): git-2.50.1-1.amzn2023.0.1.x86_64.rpm	866 kB/s 53 kB	00:00		
(2/8): perl-Error-0.17030-2.amzn2023.0.1.noarch.rpm	1.4 MB/s 42 kB	00:00		
(3/8): perl-File-Find-1.37-477.amzn2023.0.7.noarch.rpm	571 kB/s 25 kB	00:00		
(4/8): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm	1.4 MB/s 41 kB	00:00		
(5/8): perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64.rpm	1.1 MB/s 36 kB	00:00		
(6/8): perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm	539 kB/s 15 kB	00:00		
(7/8): git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm	7.2 MB/s 2.8 MB	00:00		
(8/8): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm	4.1 MB/s 4.9 MB	00:01		
Total		6.5 MB/s 7.9 MB	00:01	

```
=====
```

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
 Preparing :
 Installing : git-core-2.50.1-1.amzn2023.0.1.x86_64

3.Clone Project from Github

git clone

```
ec2-user@ip-172-31-46-88:~  
[ec2-user@ip-172-31-46-88 ~]$ sudo git clone https://github.com/iamtruptime/node-js-app-CICD.git  
Cloning into 'node-js-app-CICD'..  
remote: Enumerating objects: 49, done.  
remote: Counting objects: 100% (31/31), done.  
remote: Compressing objects: 100% (18/18), done.  
Receiving objects: 100% (49/49), 5.54 KiB | 5.54 MiB/s, done.  
remote: Total 49 (delta 20), reused 21 (delta 13), pack-reused 18 (from 1)  
Resolving deltas: 100% (24/24), done.  
[ec2-user@ip-172-31-46-88 ~]$
```

4.Go Inside Project Folder

cd nodejs.app

```
ec2-user@ip-172-31-46-88:~/node-js-app-CICD
[ec2-user@ip-172-31-46-88 ~]$ ls
node-js-app-CICD
[ec2-user@ip-172-31-46-88 ~]$ cd node-js-app-CICD/
[ec2-user@ip-172-31-46-88 node-js-app-CICD]$ |
```

5.Install Dependencies

npm install

```
ec2-user@ip-172-31-46-88:~/node-js-app-CICD
[ec2-user@ip-172-31-46-88 node-js-app-CICD]$ sudo npm install
added 68 packages, and audited 69 packages in 5s

15 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.12.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.12.1
npm notice To update run: npm install -g npm@11.12.1
npm notice
[ec2-user@ip-172-31-46-88 node-js-app-CICD]$ |
```

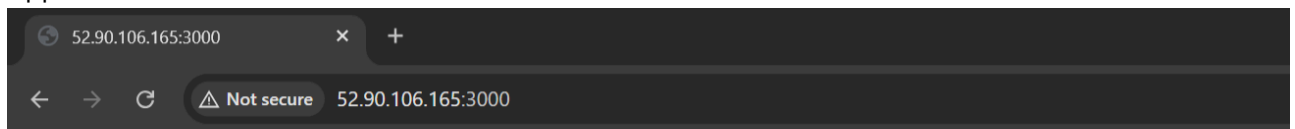
6.Run Application

node app.js

```
ec2-user@ip-172-31-46-88:~/node-js-app-CICD
[ec2-user@ip-172-31-46-88 node-js-app-CICD]$ sudo node app.js
App listening at http://localhost:3000
```

7.Access Application

- open browser
- Enter-> your-ec2-public-ip:port
- Application Will be Live



Hello from nodejs, deployed a website, we are from 21 jan batch

What I Learned

- Basics of Node.js
- Running server using Node.js
- Using npm to install dependencies

- Deploying application on EC2
- Difference between Node.js and NGINX/Apache

Future Improvements

- Use PM2 to keep app running
- Add domain name
- Enable HTTPS
- Deploy using Docker

Summery

This project helped me understand how Node.js works as a server and how to deploy applications without using traditional web servers. It is an important step in learning backend development, DevOps, and cloud deployment.